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Weight regain and cardiometabolic effects after withdrawal of semaglutide: The STEP 1 trial extension

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ABSTRACT

Background: The STEP 1 trial evaluated once-weekly subcutaneous semaglutide 2.4 mg in adults with overweight or obesity.

This off-treatment extension sought to explore changes in body weight and cardiometabolic risk factors 1 year after withdrawal.

Methods: An off-treatment extension was conducted in 327 participants from the STEP 1 trial following 68 weeks of intervention.

Participants ceased pharmacological intervention and structured lifestyle counseling at week 68 and were monitored to week 120.

Body weight, waist circumference, blood pressure, fasting glucose and glycated hemoglobin were evaluated at scheduled visits.

STEP 1 TRIAL EXTENSION — RESULTS & WEIGHT REGAIN

Primary Finding: One year after withdrawal of once-weekly semaglutide and lifestyle counse ling, participants regained roughly two-thirds of their prior weight loss.

Specifically, mean weight loss of 17.3 percent achieved at week 68 reverted by 11.6 percen tage points by week 120.

Placebo-treated individuals demonstrated a similar trajectory with a 1.9 percentage point weight regain during the unblinded extension.

Cardiometabolic Effects: Improvements in waist circumference, systolic blood pressure, and HbA1c largely reverted back toward baseline.

At week 120, waist circumference remained 4.2 cm below baseline in the semaglutide cohort, compared to 9.4 cm reduction at week 68.

Systolic blood pressure returned toward pre-trial parameters, demonstrating the necessity of sustained cardiometabolic intervention.

DISCUSSION AND CLINICAL IMPLICATIONS

These findings confirm that obesity is a chronic metabolic disease with robust biological adaptations defending adipose stores.

When pharmacotherapy ceases, compensatory neuroendocrine signaling drives appetite resumption and positive energy balance.

Permanent weight maintenance requires comprehensive long-term lifestyle support, nutrition calibration, and ongoing clinical oversight.

Study Limitations: The extension cohort comprised a subset of original trial participants who agreed to extended follow-up.

Furthermore, the trial population was predominantly white and female, meaning wider generalizability to South Asian cohorts requires further investigation.

Conclusion: Pharmacotherapy should be paired with durable habit formation, muscle preservation, and structured maintenance strategies.